

Integrative Metabolic–Vascular Model with Metformin Pharmacodynamics

State Variables

- $G(t)$: Plasma glucose concentration
- $I(t)$: Plasma insulin concentration
- $B(t)$: Functional β -cell mass
- $V(t)$: Vascular health index
- $S(t)$: Cumulative stroke risk
- $A(t)$: AMPK activation level
- $I_c(t)$: Chronic insulin exposure

Model Equations

Glucose Dynamics

$$\frac{dG}{dt} = R_a + R_{\text{hep}}(1 - \xi_A A) - U_{ii} - k_1(1 + \eta_A A) \frac{I}{R_I} G \quad (1)$$

Insulin Dynamics

$$\frac{dI}{dt} = -k_2 I + B \cdot \sigma \cdot \frac{1}{1 + \rho_A A} \cdot \frac{G^n}{G^n + G_0^n} \quad (2)$$

β -Cell Dynamics

$$\frac{dB}{dt} = B \cdot r_B \left(\frac{G}{G_{\text{norm}}} - 1 \right) - d_B B + \chi_A A B \quad (3)$$

Chronic Insulin Exposure

$$\frac{dI_c}{dt} = \epsilon(I - I_c) \quad (4)$$

Vascular Health

$$\frac{dV}{dt} = \delta(1 - V)(1 + \zeta_A A) - \alpha(G - G_{\text{norm}})_+ - \gamma I_c \quad (5)$$

Stroke Risk

$$\lambda(t) = \kappa(1 - V) + \theta(G - G_{\text{norm}})^2 \quad (6)$$

$$\frac{dS}{dt} = \lambda(t)(1 - S) \quad (7)$$

AMPK Activation

$$\frac{dA}{dt} = k_{\text{on}}D(t)(1 - A) - k_{\text{off}}A \quad (8)$$

Parameter Definitions

Parameter	Description	Units
R_a	Dietary glucose input	$\text{mmol L}^{-1} \text{min}^{-1}$
R_{hep}	Hepatic glucose production	$\text{mmol L}^{-1} \text{min}^{-1}$
U_{ii}	Insulin-independent glucose uptake	$\text{mmol L}^{-1} \text{min}^{-1}$
k_1	Insulin-mediated uptake rate	$(\text{mL } \mu\text{U}^{-1}) \text{min}^{-1}$
R_I	Insulin resistance index	dimensionless
k_2	Insulin clearance rate	min^{-1}
σ	Maximal insulin secretion rate	$\mu\text{U mL}^{-1} \text{min}^{-1}$
G_0	Half-saturation glucose constant	mmol L^{-1}
n	Hill coefficient	dimensionless
G_{norm}	Normal glucose level	mmol L^{-1}
r_B	β -cell proliferation rate	min^{-1}
d_B	β -cell loss rate	min^{-1}
χ_A	AMPK-mediated β -cell protection	min^{-1}
ϵ	Insulin averaging rate	min^{-1}
α	Glucose-induced vascular damage	min^{-1}
γ	Insulin-induced vascular damage	$(\text{mL } \mu\text{U}^{-1}) \text{min}^{-1}$
δ	Baseline vascular repair rate	min^{-1}
ζ_A	AMPK enhancement of vascular repair	dimensionless
κ	Vascular contribution to stroke hazard	min^{-1}
θ	Hyperglycaemia contribution to stroke hazard	$(\text{mmol L}^{-1})^{-2} \text{min}^{-1}$
k_{on}	AMPK activation rate by metformin	min^{-1}
k_{off}	AMPK decay rate	min^{-1}
η_A	AMPK enhancement of insulin sensitivity	dimensionless
ξ_A	AMPK suppression of hepatic glucose output	dimensionless
ρ_A	AMPK reduction of insulin demand	dimensionless
$D(t)$	Metformin dosing/adherence function	dimensionless

Table 1: Model parameters and their units